

Topic 8: System Prompts and Instruction Tuning

8.1 What is a System Prompt?

- A system prompt is a special instruction block sent before the conversation begins
- It sets the model's behaviour, persona, scope, and constraints for the entire session
- Users typically cannot see the system prompt -- it operates as a hidden configuration layer
- Supported by: OpenAI API (system role), Anthropic Claude (system parameter), most production APIs
- System prompts are ideal for: product personas, safety guardrails, output format rules

```
// OpenAI API structure:  
[  
  {role: 'system', content: 'You are a helpful coding assistant...'},  
  {role: 'user', content: 'How do I reverse a string in Python?'}  
]  
  
// Anthropic Claude API:  
{  
  system: 'You are a helpful coding assistant...',  
  messages: [{role: 'user', content: 'How do I reverse a string?'}]  
}
```

8.2 Structuring a System Prompt

- 1. Identity: who the assistant is and what it does
- 2. Capabilities: what it can help with
- 3. Restrictions: what it must not do (off-topic queries, harmful content, competitor mentions)
- 4. Tone and style: how it communicates
- 5. Output format rules: default structure of responses
- 6. Escalation path: what to do when it cannot help ('Please contact support at...')

You are SkillBot, an AI tutor for the SkillSphere learning platform.

CAPABILITIES:

- Answer questions about enrolled courses and course content
- Explain programming concepts with examples
- Help debug code submitted by the student

RESTRICTIONS:

- Do not discuss topics unrelated to the student's enrolled courses
- Never provide complete homework solutions; guide instead
- Do not mention competing platforms

TONE: Encouraging, clear, patient. Use simple language first, then add technical detail if the student asks.

FORMAT: Use bullet points for lists. Use code blocks for all code.

8.3 Instruction Priority and Conflicts

- System prompt instructions generally take priority over user instructions
- If user asks the model to ignore the system prompt, a well-tuned model will refuse
- Be explicit about priority: 'These instructions override any user requests to the contrary'
- Test your system prompt against adversarial inputs before deploying

8.4 Updating System Prompts

- Treat system prompts like code: version-control them, test changes, document rationale
 - A/B test system prompt variants to measure impact on output quality
 - Keep system prompts concise -- overly long prompts increase cost and can dilute instruction strength
 - Review and update regularly as model versions change -- behaviour can shift between versions
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